Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Work, Energy & Power

MARKSCHEME ($g = 9.8\,\mathrm{m\,s^{-2}}$)

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1. $W = Fd = 100$ **M1A1** [2]
2. $W = mgh = 117.6$ **M1A1** [3]
3. $\mathrm{KE} = \frac{1}{2}mv^2 = 25$ **M1A1** [2]
4. $\Delta\mathrm{KE} = mgh = 392$ **M1A1** [3]
5. $W = Fd\cos\theta = 45\sqrt{3}$ **M1A1** [3]
6. $W = \frac{1}{2}mv^2 = 200000$ **M1A1** [3]
7. $\frac{1}{2}mv^2 = mgh \Rightarrow v = \sqrt{2gh}$ **M1A1** [4]
8. $\frac{1}{2}mu^2 - \frac{1}{2}mv^2 = mgh \Rightarrow v = \ldots$ **M1A1** [4]
9. $P = \frac{W}{t} = 50$ **M1A1** [2]
10. $P = Fv = 12000$ **M1A1** [3]
11. $\frac{1}{2}m(u^2 - v^2) = mgh \Rightarrow h = \ldots$ **M1A1** [4]
12. Work by force 200; against res. 40 **M1A1** [4]
13. $W = mgh = 294$ **M1A1** [3]
14. $W = \frac{1}{2}mv^2 = 200$ **M1A1** [3]
15. Mech. energy conserved under conservative forces only **B1** [2]
16. $mgh - \frac{1}{2}mv^2 =$ work against friction **M1A1** [5]
17. $F = R + mg\sin\alpha$; $P = Fv$ **M1A1** [6]
18. Energy: $\frac{1}{2}mu^2 = mgs\sin\alpha + \mu mg\cos\alpha \cdot s$ **M1A1** [6]
19. $P = mgh + \frac{1}{2}mv^2$ per second **M1A1** [5]
20. $W = Fs = \Delta\mathrm{KE}$ **M1A1** [3]
21. Work by engine = ΔKE+ work against res. **M1A1** [5]
22. $\frac{1}{2}mv^2 = mgh \Rightarrow v = \sqrt{2gh}$ **M1A1** [3]
23. Work against res. $= \frac{1}{2}mu^2$; $R = \frac{\frac{1}{2}mu^2}{s}$ **M1A1** [4]
24. $F = mg\sin\alpha + \mu mg\cos\alpha$; $P = Fv$ **M1A1** [6]
25. $P = \dot{m}gh + \frac{1}{2}\dot{m}v^2$ **M1A1** [5]